

Active Reconnaissance Guide

Methods for Host Discovery (Ping Sweep) and Service Identification (Banner Grabbing)

Active reconnaissance involves directly probing target networks and services to discover live hosts, open ports, OS details, and running software versions.

1. Ping Sweeping (Host Discovery)

Ping sweeping scans IP ranges to identify active devices on a network prior to port scanning.

Method A: Nmap Host Discovery

Nmap provides versatile host discovery modes to bypass basic firewall filtering:

```
# Standard ICMP Echo Sweep (Host discovery only, disables port scan)
nmap -sn -PE 192.168.1.0/24
```

```
# TCP SYN Ping (Bypasses firewalls blocking ICMP)
nmap -sn -PS80,443 192.168.1.0/24
```

```
# ARP Scan (Best for local subnets; impossible for target to block)
nmap -sn -PR 192.168.1.0/24
```

Method B: Fast Scanning with fping

Designed to quickly process multiple target IPs simultaneously in shell scripts or CLI workflow.

```
# Sweep IP range and show only live targets (-a) while suppressing errors
fping -a -g 192.168.1.1 192.168.1.254 2>/dev/null
```

Method C: Local ARP Discovery with netdiscover

Scans or passively monitors ARP traffic on local wireless or Ethernet networks.

```
# Actively scan local network subnet for live hosts
sudo netdiscover -r 192.168.1.0/24
```

2. Banner Grabbing (Service & Version Detection)

Banner grabbing queries open ports to retrieve version tags broadcasted by services (HTTP, FTP, SSH, SMTP).

Method A: Automated Nmap Version Scanning

Interrogates open ports against thousands of known service signatures.

```
# Basic service version detection
nmap -sV 192.168.1.50
```

```
# Aggressive scan: Version detection, OS detection (-O), and default scripts (-sC)
nmap -sV -sC -O 192.168.1.50
```

Method B: Manual Connection via netcat (nc)

Directly connects to target ports to capture raw banners.

```
# Grab FTP / SSH / SMTP banner
nc -vn 192.168.1.50 21
-----
Output example: 220 (vsFTPd 3.0.3)
```

```
# Grab HTTP Header Banner manually
nc -vn 192.168.1.50 80 HEAD / HTTP/1.1 Host: 192.168.1.50
```

Method C: Quick Web Banner Probe with curl

```
# Retrieve HTTP response headers (Server, X-Powered-By, Cookies)
curl -I http://192.168.1.50
```

Method D: Web Technology Identification via whatweb

```
# Identify CMS, web server software, embedded scripts, and headers
whatweb 192.168.1.50
```

Security Tip: When performing active reconnaissance, always ensure proper authorization. Active traffic (especially SYN/Version probes) will leave direct log entries on target intrusion detection systems (IDS) and firewalls.

Quick Command Reference

Target Objective	Command	Primary Tool
Fast Live Host Ping Sweep	<code>nmap -sn 192.168.1.0/24</code>	Nmap
Local Subnet ARP Scan	<code>sudo netdiscover -r 192.168.1.0/24</code>	Netdiscover
Full Service & Version Scan	<code>nmap -sV 192.168.1.50</code>	Nmap
HTTP Header Inspection	<code>curl -I http://192.168.1.50</code>	Curl
Manual Raw Banner Probe	<code>nc -vn 192.168.1.50 <port></code>	Netcat